

# Agriculture Data Analysis & Crop Yield Insights

## 1. Introduction

Agriculture is a crucial sector that plays a vital role in food security and the economy. Understanding key factors affecting crop yield can help optimize agricultural practices. This project analyzes large agricultural datasets, investigates correlations among various features, and provides insights into crop yield trends.

## 2. Objectives

- Perform data preprocessing and exploratory data analysis (EDA).
- Identify correlations between different agricultural parameters.
- Analyze crop yield based on soil quality, rainfall, fertilizer usage, and irrigation.
- Assess water efficiency in relation to crop yield.
- Examine trends in climate change and its impact on crop yield.
- Provide recommendations for optimizing agricultural productivity based on data insights.

## 3. Dataset Overview

The dataset used in this project contains information on:

- Crop type: Categorization of different crops.
- Yield per hectare: Productivity of each crop.
- Soil pH levels: Acidity or alkalinity affecting soil quality.
- Rainfall and irrigation: Water input affecting crop growth.
- Fertilizer usage: Amount and type of fertilizers applied.
- Temperature and climatic conditions: Weather data influencing yield.
- Date and year of cultivation: Temporal aspects of crop production.

## 4. Methodology

### 4.1 Data Preprocessing

- Loaded the dataset using Pandas.
- Handled missing values using mean/mode imputation to avoid data loss.
- Converted necessary columns (e.g., date) to appropriate formats.
- Removed outliers using statistical techniques.

## 4.2 Exploratory Data Analysis (EDA)

- Generated summary statistics to understand data distribution.
- Identified correlations between features using heatmaps.
- Detected outliers using boxplots.
- Visualized crop yield distributions across different parameters.
- Performed trend analysis to examine yield changes over time.

## 4.3 Water Efficiency Analysis

- Defined water efficiency as the ratio of yield to total water input (rainfall + irrigation).
- Analyzed the impact of water efficiency on crop yield.
- Identified best-performing crops under different water conditions.

## 4.4 Climate and Seasonal Analysis

- Examined trends in yield across different months and years.
- Evaluated the effects of temperature and rainfall on crop productivity.
- Assessed climate change impact using line plots comparing yield, rainfall, and temperature over time.

## 4.5 Feature-Specific Analysis

- Studied soil pH and its impact on crop yield.
- Assessed fertilizer use efficiency and its role in yield improvement.
- Recommended best crops based on soil type and rainfall levels.

## 5. Key Findings

- Climate Impact: Rising temperatures and fluctuating rainfall patterns significantly affect crop productivity.
- Soil Quality Matters: Crops perform best in neutral pH soils (6.0-7.5), while acidic or alkaline soils reduce yield.
- Fertilizer Efficiency: Proper fertilizer application improves yield, but overuse leads to diminishing returns.
- Seasonal Variations: Yield trends fluctuate with climate conditions; certain crops perform better in dry or wet seasons.
- Water Efficiency: Optimized irrigation improves productivity, and drought-resistant crops perform better in low-rainfall areas.

## 6. Conclusion & Recommendations

### 6.1 Conclusion

- Data analysis helped identify key factors influencing crop yield.
- Trend analysis provided insights into seasonal and long-term variations.
- The study suggests that data-driven agricultural practices can significantly improve productivity.

### 6.2 Recommendations

- Precision Agriculture: Apply fertilizers and irrigation based on soil and climate conditions.
- Climate Adaptation Strategies: Use drought-resistant crops in low-rainfall regions.
- Soil Management: For acidic soils ( $\text{pH} < 5.5$ ), apply lime to neutralize acidity and improve yield.
- Water Optimization: Implement drip irrigation and rainwater harvesting to enhance water efficiency.
- Technology Integration: Develop digital dashboards for farmers to track real-time climate, soil, and crop yield insights.

## 7. Future Scope

- Expanding the dataset with real-time agricultural data from IoT sensors.
- Developing a dashboard for farmers to track climate, soil, and crop yield insights.
- Further analysis using advanced statistical models for better decision-making.